



BLM2502

Theory of

Computation

Spring 2016

BLM2502 Theory of Computation

» Course Outline

- | » Week | Content |
|-------------|---|
| » 1 | Introduction to Course |
| » 2 | Computability Theory, Complexity Theory, Automata Theory, Set Theory, Relations, Proofs, Pigeonhole Principle |
| » 3 | Regular Expressions |
| » 4 | Finite Automata |
| » 5 | Deterministic and Nondeterministic Finite Automata |
| » 6 | Epsilon Transition, Equivalence of Automata |
| » 7 | Pumping Theorem |
| » 8 | April 10 - 14 week is the first midterm week |
| » 9 | Context Free Grammars |
| » 10 | Parse Tree, Ambiguity, |
| » 11 | Pumping Theorem |
| » 12 | Turing Machines, Recognition and Computation, Church-Turing Hypothesis |
| » 13 | Turing Machines, Recognition and Computation, Church-Turing Hypothesis |
| » 14 | May 22 – 27 week is the second midterm week |
| » 15 | Review |
| » 16 | Final Exam date will be announced |



The Pumping Lemma for CFL's



Simplifications of Context-Free Grammars

A Substitution Rule

$$S \rightarrow \cancel{a}B \mid ab \mid aaA$$

$$A \rightarrow aaA \mid$$

$$A \rightarrow ab\cancel{B}c \mid$$

$$\cancel{B} \rightarrow aA$$

$$\cancel{B} \rightarrow b$$

Substitute

$abbc \mid abaAc$

$$B \rightarrow b$$

Equivalent
grammar

$$S \rightarrow \bar{a}B \mid ab$$

$$A \rightarrow aaA$$

$$A \rightarrow abBc \mid abbc$$

$$B \rightarrow aA$$



$$S \rightarrow aB \mid ab$$

$$A \rightarrow aaA$$

$$A \rightarrow abBc \mid abbc$$

$$B \rightarrow aA$$

Substitute

$$B \rightarrow aA$$

$$S \rightarrow \cancel{aB} \mid ab \mid aaA$$

$$A \rightarrow aaA$$

$$A \rightarrow \cancel{abBc} \mid abbc \mid abaAc$$

Equivalent
grammar ➤

In general: $A \rightarrow xBz$

$$B \rightarrow y_1$$

Substitute

$$B \rightarrow y_1$$

$$A \rightarrow xBz \mid xy_1z$$

equivalent
grammar ➤

Nullable Variables

ε – production :

$$X \rightarrow \varepsilon$$

Nullable Variable:

$$Y \Rightarrow \dots \Rightarrow \varepsilon$$

Example:

$$S \rightarrow aMb \mid ab$$

$$M \rightarrow aMb \mid ab$$

~~$$M \rightarrow \varepsilon$$~~

Nullable variable

ε – production



Removing ε – productions

$$S \rightarrow aMb \quad | \quad ab$$

$$M \rightarrow aMb \quad | \quad ab$$

~~$$M \rightarrow \varepsilon$$~~

Substitute

$$M \rightarrow \varepsilon$$

$$S \rightarrow aMb \mid ab$$

$$M \rightarrow aMb \mid ab$$

After we remove all the ε – productions
all the nullable variables disappear
(except for the start variable)



Unit-Productions

Unit Production: $X \rightarrow Y$

(a single variable in both sides)

Example: $S \rightarrow aA$

$A \rightarrow a$

$A \rightarrow B$

$B \rightarrow A$

$B \rightarrow bb$

Unit Productions



Removal of unit productions:

$$S \rightarrow aA \mid \cancel{aB} \mid abb \mid aa \mid a$$

$$A \rightarrow a$$

$$\cancel{A \rightarrow B}$$

$$B \rightarrow A \mid a$$

$$B \rightarrow bb$$

Substitute

$$A \rightarrow B$$

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$B \rightarrow A \mid B$$

$$B \rightarrow bb$$



Unit productions of form $X \rightarrow X$
can be removed immediately

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$B \rightarrow A \mid \cancel{B}$$

$$B \rightarrow bb$$

Remove

$$B \rightarrow B$$

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$B \rightarrow A$$

$$B \rightarrow bb$$



$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

~~$$B \rightarrow A$$~~

$$B \rightarrow bb$$

göndörün gere A gárván

Substitute

$$B \rightarrow A$$

$$S \rightarrow \cancel{aA} \mid aB \mid aA$$

$$A \rightarrow a$$

$$B \rightarrow bb$$



Remove repeated productions

$$S \rightarrow \textcircled{aA} \mid aB \mid \cancel{aA}$$

$$A \rightarrow a$$

$$B \rightarrow bb$$



Final grammar

$$S \rightarrow aA \mid aB$$

$$A \rightarrow a$$

$$\textcircled{B} \rightarrow bb$$



Useless Productions

$$S \rightarrow aSb$$

$$S \rightarrow \lambda$$

$$S \rightarrow A$$

~~$A \rightarrow aA$~~ *sonuçlanamaz* **Useless Production**

Some derivations never terminate...

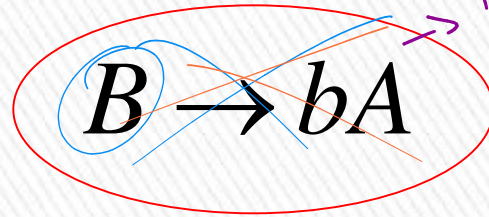
$$S \Rightarrow A \Rightarrow aA \Rightarrow aaA \Rightarrow \dots \Rightarrow aa \dots aA \Rightarrow \dots$$

Another grammar:

$$S \rightarrow A$$

$$A \rightarrow aA$$

$$A \rightarrow \lambda$$



The production $B \rightarrow bA$ is enclosed in a red oval. A blue circle highlights the non-terminal B . A blue circle highlights the non-terminal A in the right-hand side. A blue arrow points from the B in the left-hand side to the A in the right-hand side. A purple arrow points from the production to the handwritten text above it.

$$B \rightarrow bA$$

Bye his gel-mage(eh)?

Useless Production

Not reachable from S



In general:

If there is a derivation

$$S \Rightarrow \dots \Rightarrow xAy \Rightarrow \dots \Rightarrow w \in L(G)$$



consists of
terminals

Then variable A is useful

Otherwise, variable A is useless



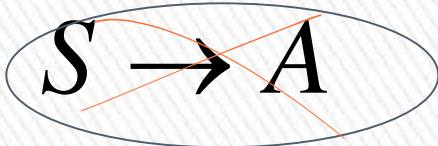
A production $A \rightarrow x$ is useless
if any of its variables is useless

$$S \rightarrow aSb$$

$$S \rightarrow \varepsilon$$

Productions

Variables


$$S \rightarrow A$$

useless

useless

$$A \rightarrow aA$$

useless

useless

$$B \rightarrow C$$

useless

useless

$$C \rightarrow D$$

useless



Removing Useless Variables and Productions

Example Grammar:

$$S \rightarrow aS \mid A \mid \cancel{C}$$

$$A \rightarrow a$$

$$\cancel{B \rightarrow aa}$$

$$\cancel{C \rightarrow aCb}$$



First: find all variables that can produce strings with only terminals or ε (possible useful variables)

$$S \rightarrow aS \mid \textcircled{A} \mid C$$

$$\textcircled{A \rightarrow a}$$

$$\textcircled{B \rightarrow aa}$$

$$C \rightarrow aCb$$

Round 1: $\{A, B\}$

(the right hand side of production that has only terminals)

Round 2: $\{A, B, S\}$

(the right hand side of a production has terminals and variables of previous round)

This process can be generalized >

Then, remove productions that use variables other than $\{A, B, S\}$

$$S \rightarrow aS \mid A \mid \cancel{C}$$

$$A \rightarrow a$$

$$B \rightarrow aa$$

$$\cancel{C \rightarrow aCb}$$



$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

$$B \rightarrow aa$$



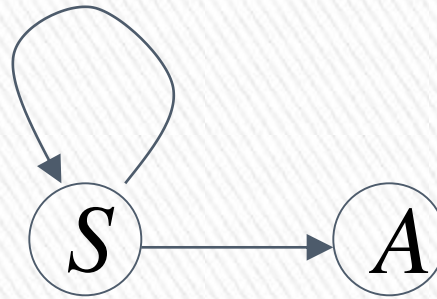
Second: Find all variables
reachable from S

Use a Dependency Graph
where nodes are variables

$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

$$B \rightarrow aa$$



unreachable



Keep only the variables
reachable from S

$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

~~$$B \rightarrow aa$$~~



Final Grammar

$$S \rightarrow aS \mid A$$

$$A \rightarrow a$$

Contains only
useful variables

Removing All

» **Step 1:** Remove Nullable Variables

$$\hookrightarrow A \rightarrow \epsilon$$

» **Step 2:** Remove Unit-Productions

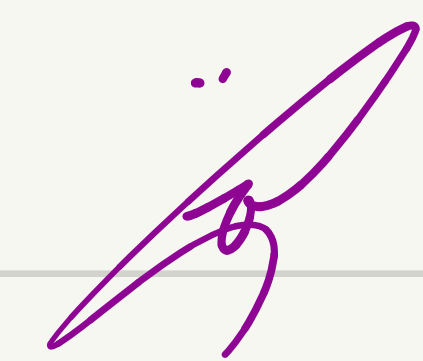
$$\hookrightarrow A \rightarrow B$$

» **Step 3:** Remove Useless Variables

$$\hookrightarrow A \rightarrow A_a$$

This sequence guarantees that unwanted variables and productions are removed





$$S \rightarrow ABA C$$

$$A \rightarrow qA \mid \varepsilon$$

$$B \rightarrow bB \mid \varepsilon$$

$$C \rightarrow c$$

① Remove null productions

$$A \rightarrow \varepsilon, B \rightarrow \varepsilon$$



$$S \rightarrow ABA C \mid ABC \mid BA C \mid BC \mid C \mid AAC \mid AC$$

$$A \rightarrow qA \mid q$$

$$B \rightarrow bB \mid b$$

$$C \rightarrow c$$

$$S \rightarrow ABA c \mid AB c \mid BA c \mid B c \mid c \mid AA c \mid A c$$

$$A \rightarrow qA \mid q$$

$$B \rightarrow bB \mid b$$

$$S \rightarrow ABA \varepsilon \mid AB \varepsilon \mid BA \varepsilon \mid B \varepsilon \mid AA \varepsilon \mid A \varepsilon \mid \varepsilon$$

$$A \rightarrow qA \mid q$$

$$B \rightarrow qB \mid b$$

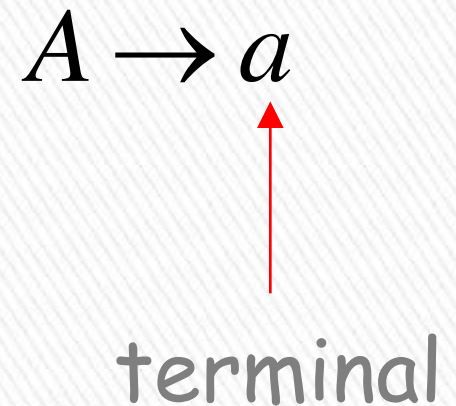
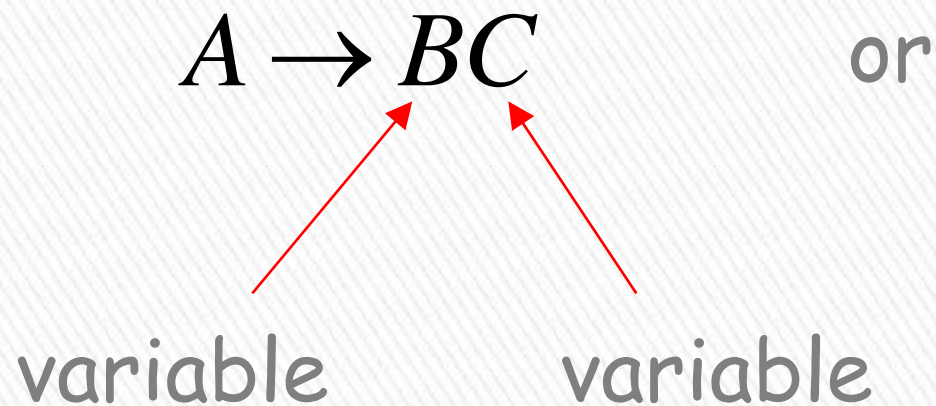
$$C \rightarrow c$$



Normal Forms for Context-free Grammars

Chomsky Normal Form

Each productions has form:



Examples:

$$\underline{S \rightarrow AS}$$

$$\underline{S \rightarrow a}$$

$$\underline{A \rightarrow SA}$$

$$\underline{A \rightarrow b}$$

$$\begin{array}{l} S \rightarrow \underline{AV_1} \\ V_1 \rightarrow AS \end{array}$$

$$T_1 \rightarrow a$$

$$S \rightarrow \underline{AS}$$

$$S \rightarrow \underline{AAS}$$

$$A \rightarrow SA$$

$$A \rightarrow \underline{aa}$$
$$A \rightarrow T_1 t_1$$

Chomsky

Normal Form

Not Chomsky

Normal Form

Conversion to Chomsky Normal Form

» Example: $S \rightarrow ABa$
 $A \rightarrow aab$
 $B \rightarrow Ac$

Not in Chomsky Normal Form

$$V_1 \rightarrow a$$

$$V_2 \rightarrow \square V_1$$

$$V_3 \rightarrow b$$

$$V_4 \rightarrow V_1 V_3$$

$$V_5 \rightarrow c$$

We will convert it to Chomsky Normal Form



Introduce new variables for the terminals:

$$T_a, T_b, T_c$$

$$S \rightarrow ABT_a$$

$$A \rightarrow T_a T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$



$$S \rightarrow ABa$$

$$A \rightarrow aab$$

$$B \rightarrow Ac$$



Handwritten blue notes:

$$\begin{aligned} T_a &\rightarrow a \\ T_b &\rightarrow b \\ T_c &\rightarrow c \end{aligned}$$

Introduce new intermediate variable V_1
to break first production:

$$S \rightarrow ABT_a$$

$$A \rightarrow T_a T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$



$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_a T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$



Introduce intermediate variable: V_2

$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_a T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$



$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_a V_2$$

$$V_2 \rightarrow T_a T_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$



Final grammar in Chomsky Normal Form:

$$S \rightarrow AV_1$$

$$V_1 \rightarrow BT_a$$

$$A \rightarrow T_aV_2$$

$$V_2 \rightarrow T_aT_b$$

$$B \rightarrow AT_c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$

Initial grammar

$$S \rightarrow ABa$$

$$A \rightarrow aab$$

$$B \rightarrow Ac$$



In general:

From any context-free grammar
(which doesn't produce ϵ)
not in Chomsky Normal Form

we can obtain:

an equivalent grammar
in Chomsky Normal Form



The Procedure

First remove:

Nullable variables

Unit productions

(Useless variables optional)



Then, for every symbol a :

New variable: T_a

Add production $T_a \rightarrow a$

In productions with length at least 2
replace a with T_a

Productions of form $A \rightarrow a$
do not need to change!



Replace any production $A \rightarrow C_1 C_2 \cdots C_n$

with $A \rightarrow C_1 V_1$

$V_1 \rightarrow C_2 V_2$

...

$V_{n-2} \rightarrow C_{n-1} C_n$

New intermediate variables: $V_1, V_2, \dots, V_{n-2}$ 

Observations

- Chomsky normal forms are good for parsing and proving theorems
- It is easy to find the Chomsky normal form for any context-free grammar



Greinbach Normal Form

All productions have form:

$$A \rightarrow a V_1 V_2 \cdots V_k \quad k \geq 0$$

symbol

variables



Examples:

$$S \rightarrow \underline{c}AB$$

$$A \rightarrow aA \mid bB \mid b$$

$$B \rightarrow b$$

Greibach
Normal Form

$\underline{b} \checkmark$

$$T_1 = b$$

$$S \rightarrow aT_1ST_1$$

$$S \rightarrow abSb$$

$$S \rightarrow \underline{aa}$$

$$T_2 = a$$

$$S \rightarrow aT_2$$

Not Greibach
Normal Form



Conversion to Greinbach Normal Form:

$$S \rightarrow abSb$$

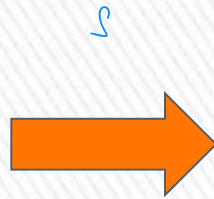
Handwritten notes: V_1 above a , V_1 above b . The a and b in $abSb$ are circled in orange.

$$S \rightarrow aa$$

Handwritten notes: V_2 above the first a , and a red slash over the second a .

$$V_1 \Rightarrow b$$

$$V_2 \rightarrow a$$



$$S \rightarrow aT_bST_b$$

$$S \rightarrow aT_a$$

Handwritten note: A blue underline under the a in aT_a .

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

Greinbach

Normal Form >

Observations

- Greinbach normal forms are very good for parsing strings (better than Chomsky Normal Forms)
- However, it is difficult to find the Greinbach normal of a grammar



BLM2502 Theory of Computation

